SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.

- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

<u>Task</u>

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

<u>Task</u>

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3

Destination Network	**Next Hop**
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Question 1

Given

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

Step 1: Identify the Subnet

A **/26** subnet has a block size of **64**.

Subnets:

- 192.168.10.0 – 63
- **192.168.10.64 – 127**
- 192.168.10.128 – 191
- 192.168.10.192 – 255

Therefore, **192.168.10.65 belongs to subnet 192.168.10.64/26.**

Step 2: Network Details

- **Network Address:** 192.168.10.64
- **Broadcast Address:** 192.168.10.127
- **Valid Host Range:** 192.168.10.65 – 192.168.10.126

Step 3: Verify Configuration

- IP Address **192.168.10.65 → Valid**
- Default Gateway **192.168.10.1 → Invalid** (belongs to a different subnet)

Step 4: Configuration Error

The default gateway is not in the same subnet as the workstation.

Step 5: Optimized Configuration

- **IP Address:** 192.168.10.65

- **Default Gateway: 192.168.10.66** (or any valid router IP within the subnet)

Step 6: Usable Hosts

$$2^{(32-26)} - 2 = 64 - 2 = 62$$

Usable Hosts = 62

Conclusion

The workstation cannot communicate because the default gateway is incorrect. After assigning a gateway within the same subnet, communication will be restored.

Question 2

Given

- **Device A:** 2001:db8:: ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Step 1: Expanded Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network Prefix

- Device A: **2001:db8:0:0::/64**
- Device B: **2001:db8:0:0::/64**

Step 3: Verify Subnet

Both devices share the same **/64 network prefix**, so they belong to the **same subnet**.

Step 4: Addressing Issue

There is **no IPv6 addressing or prefix mismatch**. The communication problem may be due to router configuration, disabled interfaces, or firewall settings.

Step 5: Optimized Configuration

Example addresses:

- Device A → **2001:db8:0:0::10/64**
- Device B → **2001:db8:0:0::20/64**

Step 6: Available Host Addresses

$$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$$

Conclusion

Both devices are correctly configured in the same subnet. The issue is likely caused by another network configuration rather than the IPv6 addresses.

Question 3

Given

Network:
192.168.1.0/24

Subnets:

- /26
- /27
- /28

Step 1: Subnet Details

Subnet	Network Address	Host Range	Broadcast	Usable Hosts
/26	192.168.1.0	192.168.1.1 – 192.168.1.62	192.168.1.63	62

Subnet	Network Address	Host Range	Broadcast	Usable Hosts
/27	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95	30
/28	192.168.1.96	192.168.1.97 – 192.168.1.110	192.168.1.111	14

Step 2: Usable and Unused Addresses

Subnet	Total Addresses	Usable Hosts
/26	64	62
/27	32	30
/28	16	14

Unused addresses depend on the number of devices connected in each department.

Step 3: Inefficient Allocation

- Large departments may run out of IP addresses if assigned a **/28**.
- Small departments waste addresses if assigned a **/26**.

Step 4: Debugging

The subnet allocation does not match the actual number of devices, leading to inefficient IP usage.

Step 5: Optimized Solution (VLSM)

Use **Variable Length Subnet Masking (VLSM)**:

- Large Department → **/26**
- Medium Department → **/27**
- Small Department → **/28**

This reduces address wastage and improves network efficiency.

Conclusion

VLSM provides a better subnet allocation by assigning subnet sizes according to departmental requirements.

Question 4

Given

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Step 1: Expanded Addresses

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network Prefix

Both devices have the network prefix:

2001:db8:0:0::/64

Step 3: Verify Subnet

Both devices belong to the **same IPv6 subnet**.

Step 4: Addressing Issue

There is **no IPv6 address mismatch**. The communication problem may be caused by incorrect routing, firewall settings, or disabled interfaces.

Step 5: Optimized Configuration

Assign valid IPv6 addresses such as:

- Device A → **2001:db8:0:0::10/64**
- Device B → **2001:db8:0:0::20/64**

Step 6: Available Host Addresses

$$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$$

Conclusion

The IPv6 addresses are correctly configured. Network devices and routing should be checked to resolve the communication issue.

Question 5

Given Routing Table

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP:
192.168.2.100

Step 1: Longest Prefix Match

The destination IP belongs to:

192.168.2.0/24

Step 2: Matching Route

The router selects:

192.168.2.0/24 → 10.0.0.3

Step 3: Next-Hop Router

10.0.0.3

Step 4: Verify Destination

The IP **192.168.2.100** falls within the valid host range of the **192.168.2.0/24** subnet.

Step 5: Routing Error

Possible reasons for failure:

- Incorrect next-hop IP address

- Interface is down
- Missing return route
- Incorrect static route configuration

Step 6: Optimized Routing

- Verify the next-hop **10.0.0.3**
- Ensure router interfaces are active
- Check and add any missing routes

Step 7: Default Route

If no matching route exists, the router forwards the packet using:

0.0.0.0/0 → 10.0.0.1

Conclusion

The router correctly selects the **192.168.2.0/24** route and forwards packets to **10.0.0.3**. If no matching destination is found, it uses the default route through **10.0.0.1**.